MRVL Marvell Technology, Inc. — 个股深度研究 (专业中文版) v20261016-0756

> 数据截至 2026-10-16 07:56 SGT｜价格 $241.24｜市值 216.8 bn USD｜财报货币 USD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

Marvell 受益于 AI 光互连(PAM4 DSP)与定制 XPU；争议点在定制项目的代际延续性及光 DSP 是否被 CPO 侵蚀。

- **多头驱动**: Next-gen Trainium/Maia sockets retained; 1.6T DSP ramp｜下一代 Trainium/Maia 留住；1.6T DSP 放量
- **空头驱动**: Losing socket to rival; LPO/CPO shrinks DSP TAM｜失去关键项目；LPO/CPO 压缩 DSP 市场
- **证伪条件 Falsifiers**: DC revenue guide flat-to-down; customer disclosures of competitor sockets｜DC 收入指引持平或下降；客户披露换供应商

2. 产品、客户与收入模式

- 主题 Theme: Custom ASIC + electro-optics 定制 ASIC+光电 DSP
- 业务分部 Segments: Data center (custom+optical DSP) 数据中心；Enterprise/Carrier 企业/运营商；Consumer/Auto/Industrial 其他
- 公司描述 (DATA, Yahoo/公司申报): Marvell Technology, Inc., together with its subsidiaries, provides data infrastructure semiconductor solutions and spanning the data center core to network edge in the United States, Argentina, China, India, Israel, Japan, Singapore, South Korea, Taiwan, Vietnam, and internationally. The company develops and scales system-on-a-chip architectures, integrating analog, mixed-signal, and digital signal processing functionality. It offers a portfolio of ethernet solutions, including spanning controllers, network adapters, physical transceivers, and switches; single or multiple core processors; and custom application specific integrated circuits, interconnects, fibre channel adapters, and processors. The company also provides interconnect products, including pulse amplitude modulation, coherent and coherent-lite digital signal processors (DSPs), laser drivers, trans-impedance amplifiers, silic
- 主要客户: AWS (Trainium), Microsoft (Maia), Google, hyperscaler optical module makers｜AWS、微软、谷歌、光模块厂

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: TSMC, substrate/packaging vendors｜台积电、载板/封装
- 竞争者 (含业外): Broadcom, Alchip, MediaTek, Credo (AECs), Inphi-legacy rivals｜博通、世芯、联发科、Credo
- 瓶颈/监管: Custom program wins/losses, PAM4 DSP vs LPO/CPO substitution｜定制项目得失、DSP 被 LPO/CPO 替代
- 关键 KPI: DC revenue, custom revenue, optical DSP revenue, GM%｜DC 收入、定制收入、光 DSP 收入、毛利率
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, USD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-07-31	2.7 bn USD	1.5 bn USD	0.5 bn USD	0.3 bn USD	0.5 bn USD	16.7%
2026-04-30	2.4 bn USD	1.3 bn USD	0.4 bn USD	34.5 mn USD	0.5 bn USD	14.5%
2026-01-31	2.2 bn USD	1.1 bn USD	0.4 bn USD	0.4 bn USD	0.3 bn USD	18.7%
2025-10-31	2.1 bn USD	1.1 bn USD	0.4 bn USD	1.9 bn USD	0.5 bn USD	17.7%
2025-07-31	2.0 bn USD	1.0 bn USD	0.3 bn USD	0.2 bn USD	0.4 bn USD	14.9%
2025-01-31	N/A	N/A	N/A	N/A	N/A	N/A

资产负债: 现金 3.9 bn USD；总负债 5.3 bn USD；净现金 -1.4 bn USD。TTM FCF 1.7 bn USD（FCF 率 18.2%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-10-31	0.76	0.73778	0.0301
2026-01-31	0.8	0.79165	0.0105
2026-04-30	0.8	0.79495	0.0064
2026-07-31	0.94	0.93077	0.0099

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-12-02**；本季 EPS 一致预期 1.10094（34 位分析师）；收入一致预期 3.2 bn USD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$43.07	-82.1%
2Q	Base	$110.70	-54.1%
2Q	Bull	$195.60	-18.9%
4Q	Bear	$47.82	-80.2%
4Q	Base	$140.58	-41.7%
4Q	Bull	$266.79	10.6%
6Q	Bear	$53.00	-78.0%
6Q	Base	$177.90	-26.3%
6Q	Bull	$362.08	50.1%
8Q	Bear	$58.66	-75.7%
8Q	Base	$224.44	-7.0%
8Q	Bull	$489.28	102.8%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	50.9x
4Q	37.8x
6Q	28.1x
8Q	21.0x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	31.6%	24.0%	14.2%	2.5%	25.0%	$60.32	-75.0%
Base	63.2%	30.0%	13.2%	3.0%	50.0%	$257.58	6.8%
Bull	82.2%	34.5%	12.7%	3.5%	25.0%	$601.20	149.2%

概率加权价值 **$294.17**（相对现价 21.9%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 232.8 bn USD；终值占比 56.0%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **61.2%**；一致预期 FY+1 增速 63.2%。
- WACC (CAPM) = rf 5.2% + β 1.60 × ERP 5.0% = 13.2%。
- SOTP segments 分部: Data center (custom+optical DSP) 数据中心; Enterprise/Carrier 企业/运营商; Consumer/Auto/Industrial 其他. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | MRVL | 216.8 bn USD | 22.5x | 33.1x | 36.5% | 52.2% | 0.8% | | AVGO | 1,643.7 bn USD | 18.8x | 17.8x | 85.5% | 75.5% | 1.9% | | CRDO | 33.5 bn USD | 20.6x | 18.4x | 114.7% | 67.1% | 0.7% | | ALAB | 52.4 bn USD | 42.6x | 47.2x | 104.5% | 75.1% | 0.2% | | AMD | 917.8 bn USD | 22.0x | 35.7x | 50.1% | 55.7% | 1.0% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: Losing socket to rival; LPO/CPO shrinks DSP TAM ｜失去关键项目；LPO/CPO 压缩 DSP 市场
- 证伪指标: DC revenue guide flat-to-down; customer disclosures of competitor sockets｜DC 收入指引持平或下降；客户披露换供应商
- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	241.2	Yahoo Finance quote, fetched 2026-10-16T07:56 SGT
Market cap 市值	2.168e+11	Yahoo info.marketCap 2026-10-16T07:56

Item	Value	Source/time
Cash 现金	3.933e+09	Yahoo info.totalCash (latest filing) 2026-10-16T07:56
Debt 负债	5.286e+09	Yahoo info.totalDebt (latest filing) 2026-10-16T07:56
Revenue TTM 收入	9.45e+09	Yahoo info.totalRevenue (TTM, USD) 2026-10-16T07:56
FCF TTM 自由现金流	1.723e+09	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-07-31', '2026-04-30', '2026-01-31', '2025-10-31'] (USD)
Beta	2.26	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.0519	FRED DGS10 2026-10-14
FX USD per USD	1	n/a (USD)
Consensus rev growth FY0	0.4729	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.6321	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market ERP
Beta clamp	[0.8,1.6] → 1.60	ASSUMPTION: limit extreme betas
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%
Year-1 growth (Base)	0.6321	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION

Assumption	Value	Rationale
Starting FCF margin	0.1823	DATA: FCF TTM / Revenue TTM, floored -50%
Long-run FCF margin (Base)	0.3	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

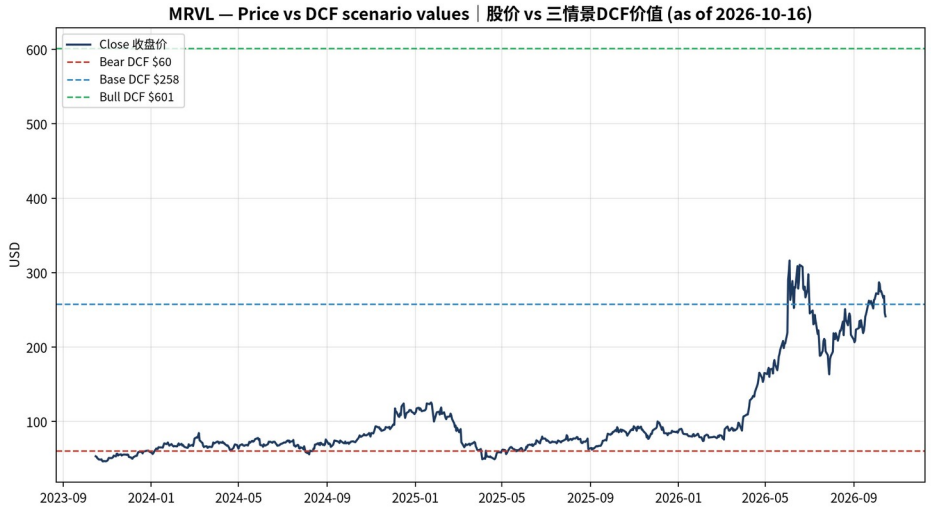
12. Model Provenance 模型溯源

- Model file: `MRVL_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT
- Version: v20261016-0756. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

- v20261016-0756: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-12-02 (TENTATIVE)。

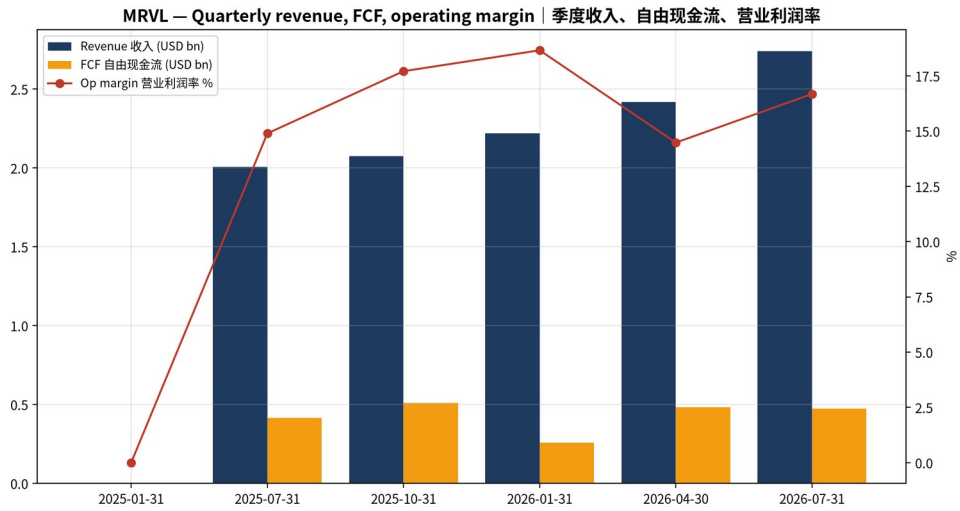
图片报告 Image Report



Claim class 声明类别: DATA (price) + HYPOTHESIS (scenario values) | Source 来源: Yahoo Finance daily closes; model MRVL_model.xlsx
Thinking 思考: Gap between price and Base line = what the market believes beyond our base assumptions → see Reverse DCF. | Not investment advice 非投资建议

1_price_vs_value

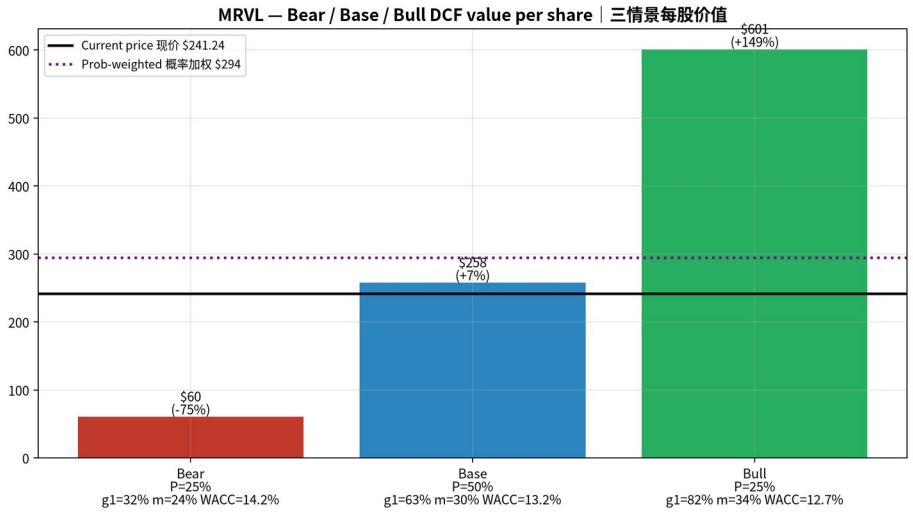
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

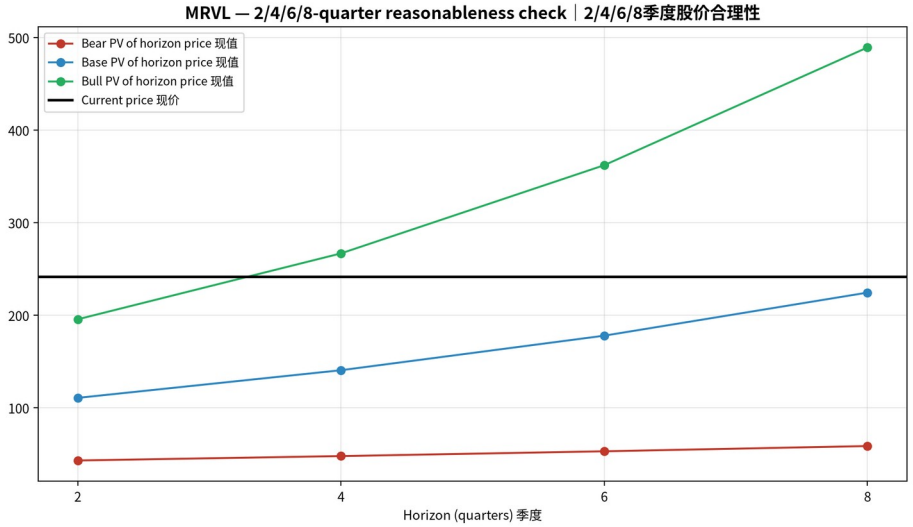
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

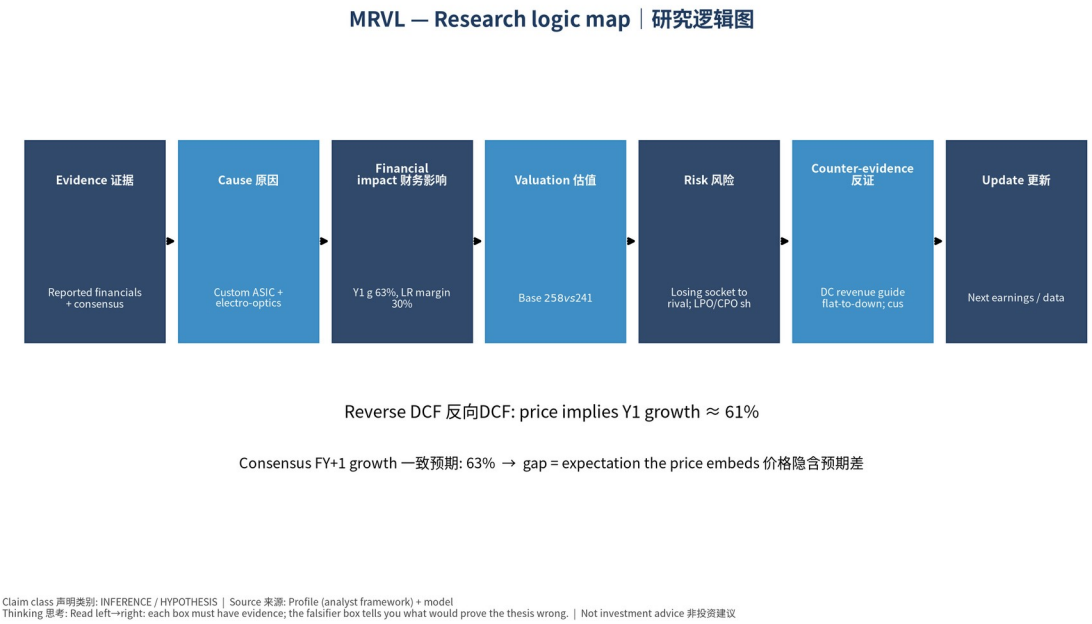
三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验 Horizon check



5_logic_map

研究逻辑图Logic map

MRVL Marvell Technology, Inc. — Equity Deep Dive (Professional English) v20261016-0756

> Data as of 2026-10-16 07:56 SGT | Price $241.24 | Mkt cap 216.8 bn USD | Reporting currency USD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

Marvell benefits from AI optical interconnect (PAM4 DSPs) and custom XPUs; the debate is generational continuity of custom programs and whether CPO erodes DSP demand.

- **Bull drivers**: Next-gen Trainium/Maia sockets retained; 1.6T DSP ramp｜下一代 Trainium/Maia 留住；1.6T DSP 放量
- **Bear drivers**: Losing socket to rival; LPO/CPO shrinks DSP TAM｜失去关键项目；LPO/CPO 压缩 DSP 市场
- **Falsifiers**: DC revenue guide flat-to-down; customer disclosures of competitor sockets｜DC 收入指引持平或下降；客户披露换供应商

2. Products, customers, revenue model

- Theme: Custom ASIC + electro-optics 定制 ASIC+光电 DSP | Segments: Data center (custom+optical DSP) 数据中心; Enterprise/Carrier 企业/运营商; Consumer/Auto/Industrial 其他
- Business description (DATA): Marvell Technology, Inc., together with its subsidiaries, provides data infrastructure semiconductor solutions and spanning the data center core to network edge in the United States, Argentina, China, India, Israel, Japan, Singapore, South Korea, Taiwan, Vietnam, and internationally. The company develops and scales system-on-a-chip architectures, integrating analog, mixed-signal, and digital signal processing functionality. It offers a portfolio of ethernet solutions, including spanning controllers, network adapters, physical transceivers, and switches; single or multiple core processors; and custom application specific integrated circuits, interconnects, fibre channel adapters, and processors. The company also provides interconnect products, including pulse amplitude modulation, coherent and coherent-lite digital signal processors (DSPs), laser drivers, trans-impedance amplifiers, silic
- Customers: AWS (Trainium), Microsoft (Maia), Google, hyperscaler optical module makers｜AWS、微软、谷歌、光模块厂

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: TSMC, substrate/packaging vendors｜台积电、载板/封装
- Competitors: Broadcom, Alchip, MediaTek, Credo (AECs), Inphi-legacy rivals｜博通、世芯、联发科、Credo
- Bottlenecks/regulation: Custom program wins/losses, PAM4 DSP vs LPO/CPO substitution｜定制项目得失、DSP 被 LPO/CPO 替代
- KPIs: DC revenue, custom revenue, optical DSP revenue, GM%｜DC 收入、定制收入、光 DSP 收入、毛利率
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, USD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-07-31	2.7 bn USD	1.5 bn USD	0.5 bn USD	0.3 bn USD	0.5 bn USD	16.7%
2026-04-	2.4 bn	1.3 bn	0.4 bn	34.5 mn	0.5 bn	14.5%

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
30	USD	USD	USD	USD	USD	
2026-01-31	2.2 bn USD	1.1 bn USD	0.4 bn USD	0.4 bn USD	0.3 bn USD	18.7%
2025-10-31	2.1 bn USD	1.1 bn USD	0.4 bn USD	1.9 bn USD	0.5 bn USD	17.7%
2025-07-31	2.0 bn USD	1.0 bn USD	0.3 bn USD	0.2 bn USD	0.4 bn USD	14.9%
2025-01-31	N/A	N/A	N/A	N/A	N/A	N/A

Cash 3.9 bn USD; debt 5.3 bn USD; TTM FCF 1.7 bn USD (FCF margin 18.2%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-10-31	0.76	0.73778	0.0301
2026-01-31	0.8	0.79165	0.0105
2026-04-30	0.8	0.79495	0.0064
2026-07-31	0.94	0.93077	0.0099

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-12-02**; current-quarter EPS consensus 1.10094 (34 analysts); revenue consensus 3.2 bn USD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$43.07	-82.1%
2Q	Base	$110.70	-54.1%
2Q	Bull	$195.60	-18.9%
4Q	Bear	$47.82	-80.2%
4Q	Base	$140.58	-41.7%
4Q	Bull	$266.79	10.6%
6Q	Bear	$53.00	-78.0%
6Q	Base	$177.90	-26.3%
6Q	Bull	$362.08	50.1%

Horizon	Scenario	PV of horizon price	vs current
8Q	Bear	$58.66	-75.7%
8Q	Base	$224.44	-7.0%
8Q	Bull	$489.28	102.8%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	50.9x
4Q	37.8x
6Q	28.1x
8Q	21.0x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/share	vs price
Bear	31.6%	24.0%	14.2%	2.5%	25.0%	$60.32	-75.0%
Base	63.2%	30.0%	13.2%	3.0%	50.0%	$257.58	6.8%
Bull	82.2%	34.5%	12.7%	3.5%	25.0%	$601.20	149.2%

Probability-weighted value **$294.17** (21.9% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 232.8 bn USD; terminal share 56.0%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **61.2%** vs consensus FY+1 63.2%.
- WACC (CAPM) = rf 5.2% + β 1.60 × ERP 5.0% = 13.2%.
- SOTP segments: Data center (custom+optical DSP) 数据中心; Enterprise/Carrier 企业/运营商; Consumer/Auto/Industrial 其他 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

Losing socket to rival; LPO/CPO shrinks DSP TAM｜失去关键项目；LPO/CPO 压缩 DSP 市场 | Falsifiers: DC revenue guide flat-to-down; customer disclosures of competitor sockets｜DC 收入指引持平或下降；客户披露换供应商

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `MRVL_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT - Version: v20261016-0756. Prior versions are never overwritten (see company history page).